Finding the Nuclei That Models Miss

Reliable instance segmentation and classification
across diverse tissues in PanNuke

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Project 5

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Project summary

Measuring tissue composition requires finding individual nuclei, separating neighboring cells, and assigning each nucleus a cell type. A model can score well overall while overlooking small, faint, or partially visible nuclei. In our PanNuke experiments, the Dead class, which includes apoptotic and necrotic nuclei, is especially difficult. We want to recover these objects while preserving mask quality and controlling false detections.

This project connects error analysis with a focused model-design question. We will consolidate our baseline results, complete a controlled study of supervision for cropped nuclei, and investigate PM-Net, a model that predicts a separate mask and type for each proposed nucleus. PM-Net replaces horizontal/vertical distance maps and watershed decoding with point-based instance queries. Matched comparisons will show whether this change is worth its added complexity.

Intellectual merit

We ask whether representing nuclei as individual objects, including those only partly visible, can improve rare-class detection while maintaining segmentation and classification quality. The study will test which errors respond to changes in supervision and which require a different instance decoder. A well-supported negative result will also help define the limits of these approaches.

Broader impacts

Reliable nucleus-level measurements support research on tissue organization and the tumor microenvironment. By sharing evaluation tools, error maps, and documented failure cases, we aim to help researchers judge improvements by the cells they recover, the mistakes they introduce, and the computation they require.

1. Problem, specific aims, and prior work

Hematoxylin and eosin (H&E) images show cells within their tissue context. Automated analysis must distinguish adjacent nuclei despite changes in staining, size, shape, and tissue architecture. Small apoptotic or necrotic structures are easily missed; crowded nuclei may merge; and inflammatory and connective nuclei can look similar. These errors can bias cell counts and estimates of tissue composition. PanNuke provides a demanding setting in which to study them [1].

Specific aims

Aim 1 - Identify the errors that matter. Bring together our HoVer-Net and CellViT-UNI results under one evaluator. Measure missed nuclei, merged neighbors, fragmented predictions, and typing errors by tissue and class. This analysis will provide the reference for each proposed change.

Aim 2 - Test supervision for cropped nuclei. Complete the ongoing four-arm, two-seed experiment using its predefined protocol. Compare stronger cropping alone with supervision that preserves horizontal/vertical (HV) targets for small fragments newly cut by a crop. Evaluate both controlled crop windows and the original validation images.

Aim 3 - Test a direct instance model. Build a small PM-Net prototype that starts from proposed nucleus locations and jointly predicts masks and five cell types. Determine whether it improves Dead detection and separates touching nuclei at a practical inference cost. Predefined development criteria will determine whether a larger confirmation study is warranted.

Prior work and the proposed difference

HoVer-Net predicts nuclear foreground, horizontal/vertical offsets, and cell type; instance separation uses the offset maps and post-processing [2]. CellViT introduces a transformer encoder into nuclei segmentation and classification [3]. Both are established baselines for this project. Our reproduced CellViT-UNI is the primary performance reference, while HoVer-Net anchors the course reproduction requirement.

Several methods suggest ways to change the instance decoder. PromptNucSeg learns a point prompt for each nucleus and uses a prompt-conditioned decoder to predict its mask [4]. Mask2Former represents objects with queries that predict masks [5]. CellVTA uses a CNN adapter and cross-attention to preserve fine spatial information [6]. Together, these approaches motivate the components of PM-Net, whose performance on our task remains to be tested.

PM-Net will reuse a pretrained encoder and our existing data and evaluation tools. The contribution we will test is how point proposals, spatial detail, and joint mask/type prediction work together on difficult nuclei. We will begin with the same UNI initialization as the baseline so that the first comparison isolates changes to the decoder.

2. Data and preliminary evidence

The main dataset is PanNuke: approximately 7,900 RGB patches at 256 x 256 pixels, spanning 19 tissue types and roughly 200,000 annotated nuclei [1]. Each nucleus has an instance mask and one of five labels: neoplastic, inflammatory, connective, dead, or epithelial. The model input is the image; the required output is a distinct mask and one class per detected nucleus. Tissue labels support stratified analysis.

The prepared folds contain 2,656, 2,523, and 2,722 patches. The current development round trains on fold 1 and evaluates on fold 2. Staining, cell density, class frequency, and tissue context vary, and many nuclei are cut by patch boundaries. Dead nuclei are scarce, making their scores sensitive to tissue composition and training seed. The annotations are a useful reference, but some visible objects and type boundaries remain uncertain.

What has already been established

Own training; historical three-split means	mPQ	bPQ
HoVer-Net, final checkpoint, no TTA	0.4564	0.6613
CellViT-UNI, final checkpoint, no TTA	0.4995	0.6653

These results come from the existing project record [7]. Our shared evaluator has been checked against the official PanNuke implementation. An earlier analysis found that 38.5% of labeled Dead nuclei had an intersection-over-union (IoU) below 0.1 with every predicted object. This substantial detection deficit motivates studying where and why nuclei disappear from the predictions.

Our original approach explored language-based reclassification, failure-driven copy-paste and image synthesis, and controlled changes to tissue appearance. The tested reclassification and augmentation recipes did not produce a dependable improvement. Synthetic appearance tests revealed typing weaknesses but did not reproduce detection rankings on real external datasets. Later experiments with classifiers on frozen features and learned candidate selection also failed their predefined criteria; these routes are now closed [7].

Combining filtered predictions from two scales retained a small gain in Dead panoptic quality (PQ), at about 6.1 times the cost of single-scale inference. Controlled crop experiments then linked partial visibility to missed nuclei. Aim 2 tests one possible explanation: small cropped fragments lose an HV training signal that may help the model recognize them.

Annotation and generalization limits

Agreement among several detectors has identified objects and type boundaries that deserve review. Those detectors can share the same mistakes, and expert review is currently unavailable. We will preserve this uncertainty in our interpretation. CoNIC, MoNuSAC, and PUMA offer secondary external checks after documenting class mappings and excluding known overlaps; they will be kept separate from method development.

3. Aim 2: a controlled supervision experiment

A training crop can leave only a small piece of a nucleus. In the existing target-generation rule, objects smaller than 30 working pixels lose the usual HV separation signal. We will test whether retaining this signal for newly cut fragments helps detection. The intervention and its controls use the same cropped images and all other labels.

Arm	Crop probability	HV target rule
A0: retrained baseline	0.1	Existing minimum area of 30
A1: crop-only control	0.5	Existing minimum area of 30
A2: selective supervision	0.5	Minimum 1 only for newly truncated objects with visible area below 30
A3: small-object control	0.5	Minimum 1 for every object; actual target differences occur below area 30

The four arms use seeds 19 and 1, 130 epochs, the same initialization, optimizer, loss weights, sampling, and decoder, and final-checkpoint evaluation at native resolution without test-time augmentation (TTA). A1-A3 share the same augmentation random sequence. A2 tracks instance identity before and after cropping; objects already cut by the original patch boundary are not treated as newly truncated. Area is measured after geometric transformations in training coordinates [8].

Current progress, checked October 14 at 8:34 p.m. Eastern

At this checkpoint, six of eight runs had completed training and fold-2 evaluation: A0-A3 for seed 19, plus A0 and A1 for seed 1. A2 seed 1 was training and A3 seed 1 was pending. For seed 19, A2 changes mPQ by -0.0091, bPQ by -0.0062, Dead PQ by +0.0222, and strict Dead PQ by +0.0025 relative to A0. Both overall-quality losses exceed the predefined limit of 0.002. Thus the full success rule cannot be met, even though the controlled-crop analysis is still needed to understand the effect on truncated nuclei [8].

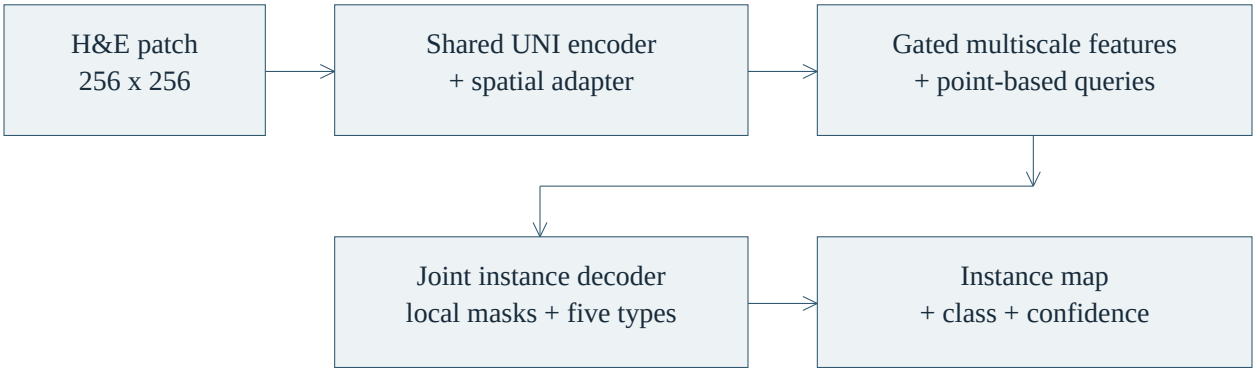
Remaining analysis and fixed decision rule

After training finishes, evaluate each model on the same 2,523 fixed crop windows, called C2 in the registered protocol. A2-A1 tests the supervision change; A2-A0 measures its overall cost and benefit; A3 tests whether lowering the cutoff for all small objects gives the same effect. The predefined rule requires fewer newly truncated Dead misses in both seeds, a mean reduction of at least 0.03, and a positive 95% lower confidence bound for improvement. It also requires no strict Dead decline versus A0 in either seed, mPQ/bPQ losses of at most 0.002 per seed, mean Dead PQ improvement of at least 0.005, and no mean strict Dead decline for A2-A1 [8].

We will keep these thresholds unchanged and report the failed quality criterion alongside the remaining mechanism analysis. Any partial-visibility task in PM-Net will need its own test; the present experiment does not establish that it improves performance.

4. Aim 3: PM-Net, a direct instance model

PM-Net is our proposed alternative to the current foreground/HV/type decoder [9]. It starts from candidate nucleus locations and predicts a mask and type for each object. The model has not yet been implemented or trained. We will begin with a small version whose components can be tested separately.



Proposed PM-Net. One shared image encoding; no HV maps or watershed.

Computational steps

1. Preserve detail. Encode each patch once with UNI. A compact CNN adapter will retain fine spatial information, and learned gates will combine features at different scales. Start with 256 x 256 images and the existing encoder initialization. Larger backbones and 512-pixel inputs can be considered after this first decoder comparison.

2. Propose nucleus locations. Predict a heatmap of likely nucleus locations and types. Each candidate point initializes an instance query, a feature vector representing one possible nucleus. Training points will lie inside the visible instance masks. Set the query limit using training-set object counts and a safety margin, then report any overflow. About 256 queries is an initial design choice to be checked.

3. Predict masks and types together. Each query uses local features and surrounding tissue context to predict a mask, confidence, and five-class distribution. During training, one-to-one matching pairs predictions with labeled nuclei using location, class, and mask agreement. Point focal loss, mask cross-entropy/Dice losses, and class cross-entropy will supervise the shared features. Select their weights on development data and fix them before confirmation. The point head receives direct supervision because discrete point selection itself need not pass gradients.

4. Produce an instance map. Assign overlapping pixels to individual masks using a confidence rule selected on validation data. Return masks to native image coordinates before scoring. Check which tiny objects are removed by any size filter; the decoder filter and the HV training cutoff in Aim 2 are different choices.

Keep the first test interpretable

Add point initialization and gated feature fusion separately to the basic query decoder. Compare with an equally sized plain adapter to check whether extra capacity explains any gain. Once the core model works, optional studies can test truncation supervision, an auxiliary label that groups inflammatory/connective nuclei, or extra training queries. The main study will retain five-class evaluation. Alternative mask decoders, larger encoders, and teacher models remain outside its minimum scope [9].

5. Experiments and evaluation

Development and confirmation

Train on fold 1 and develop on fold 2, pairing each method with the same seeds, 19 and 1. Fix the architecture, augmentation, losses, decoder, and success criteria before confirmation. Begin with the existing 130-epoch schedule, including encoder unfreezing at epoch 25. A short runtime check will establish feasibility. Use the same pretrained encoder initialization for matched runs; initialize new task heads independently. Released checkpoints trained on all PanNuke folds are ineligible for test comparisons.

If development supports further study, use the project three-split protocol: train/validation/test folds 1/2/3, 2/1/3, and 3/2/1. Evaluate final checkpoints and show every split and seed before reporting their mean. Splits 1 and 2 share test fold 3. We will preserve the released folds and disclose that available patient/slide identifiers do not allow us to verify patient-level separation.

Earlier project stages have already examined the test folds, and fold 2 has been used repeatedly for development. This limits how strongly a later confirmation can establish generalization. We will lock the method before that analysis, make no changes based on test results, and use external datasets as a secondary check.

Endpoints, comparisons, and uncertainty

Use the shared PanNuke evaluator for every method. Panoptic quality combines correct object matching with mask overlap. Report official mPQ, which includes cell type, bPQ, which ignores type, and PQ for each class. Also report strict mPQ and strict Dead PQ: these retain penalties for predicted classes absent from an image. Separate detection from typing, compare interior and border Dead misses, count merges and splits, and plot tissue-by-class performance. Show Dead results with and without Uterus to reveal tissue-specific effects.

Compare PM-Net with CellViT-UNI trained using the same recipe and with the crop-only control. Estimate paired differences by resampling images within tissues, keeping crop windows from the same patch together. Show each seed and its confidence interval. These intervals describe image-sampling variation for the trained models; patient-level uncertainty and variation across all possible training seeds remain outside their scope.

Proposed PM-Net development criteria

We propose the following criteria, to be fixed before prototype training: mean interior Dead recall improves by at least 0.03 and mean Dead PQ by at least 0.005, with both changes positive in each seed. In either seed, mPQ and bPQ may fall by at most 0.002, while strict Dead PQ and strict mPQ must not fall. Mean merge/split rates must not increase. The paired 95% lower confidence bound for the primary recall improvement must exceed zero. These proposed criteria apply only to the new model study.

The inference-cost target is at most twice the matched single-scale baseline, including decoding, on the same hardware and batch size. Measure memory and latency repeatedly and record competing device load. Failure on quality or cost will stop expansion. Further work would require a revised plan with a stated reason for changing course.

6. Scope, milestones, risks, and broader impact

Window from proposal agreement	Work and concrete output
Week 1	Close the visibility round; complete fixed-window analysis; consolidate baseline and failure tables.
Week 2	Freeze PM-Net design and admission rule; implement a small prototype; check masks, matching, query capacity, memory, and timing.
Weeks 3-4	Run the bounded two-seed development comparison and core ablations, subject to resource availability. Publish a go/no-go analysis.
Weeks 5-6	If warranted, conduct separately approved three-split confirmation and external checks. Otherwise finish the negative-result analysis and final report.

The schedule is relative to proposal agreement. **The minimum result** is the reproduced baselines, the completed supervision experiment, and an explanation of error and cost tradeoffs. **The target result** adds a controlled PM-Net prototype. **Stretch goals** are three-split confirmation, external evaluation, and one justified auxiliary task.

Feasibility and risks

Rare labels and uncertain annotation. Report class counts, tissue breakdowns, and paired uncertainty. Preserve original labels. Detector agreement can identify examples for discussion; expert review is still needed to judge their biological identity.

Better detection but poor masks. A point can locate a nucleus whose predicted mask still fails the overlap criterion. Examine point recall, mask quality, duplicates, and crowded-object errors separately. If the query model cannot produce useful masks, stop its expansion and complete the baseline and failure-analysis deliverables.

Confounding and scope growth. Hold the encoder and recipe fixed while testing gates, points, and auxiliary supervision separately. Keep closed retyping and synthesis routes closed. Discuss substantive changes to the aims with mentors and record the reasons.

Compute, storage, and interruption. Estimate 2-4 A100 GPU-days for the prototype and a further 4-8 for confirmation, subject to allocation. Storage is tight: verify capacity and agree on archiving before training. Validate resumable state for future runs before freezing them; preserve current checkpoints.

Deliverables and broader impact

Deliver versioned evaluation code, configurations, model and data provenance, final models, comparison tables, annotated failure examples, tissue/class heatmaps, and a final report. Success may take the form of a repeatable improvement that meets the quality and cost criteria, or a clear explanation of why the intervention fails. Either outcome can guide more reliable cell measurement and help other groups avoid costly experiments that simply move errors between metrics. Clinical use would require further validation and expert review.

References and project evidence

[1] Gamper, J., et al. (2020). PanNuke Dataset Extension, Insights and Baselines.
https://arxiv.org/abs/2003.10778

[2] Graham, S., et al. (2019). HoVer-Net: Simultaneous segmentation and classification of nuclei in multi-tissue histology images. Medical Image Analysis, 58, 101563.
https://doi.org/10.1016/j.media.2019.101563

[3] Hörst, F., et al. CellViT: Vision Transformers for Precise Cell Segmentation and Classification. arXiv:2306.15350.
https://arxiv.org/abs/2306.15350

[4] Shui, Z., et al. Unleashing the Power of Prompt-driven Nucleus Instance Segmentation. arXiv:2311.15939.
https://arxiv.org/abs/2311.15939

[5] Cheng, B., et al. (2022). Masked-attention Mask Transformer for Universal Image Segmentation. CVPR.
https://arxiv.org/abs/2112.01527

[6] Yang, Y., et al. (2025). CellVTA: Enhancing Vision Foundation Models for Accurate Cell Segmentation and Classification. arXiv:2504.00784.
https://arxiv.org/abs/2504.00784

[7] bdd26 project records. README; RESEARCH_PLAN; research closeout audit; stage reports I-V. Repository snapshot b60c776 (October 14, 2026). Historical results cited here were consolidated from these records; they were not all recomputed for this proposal.

Versioned project documentation

[8] bdd26. Visibility-supervision frozen specification (October 14, 2026); scientific training source eee3248. Current progress and six original-image fold-2 summary files inspected October 14, 2026, 8:34 p.m. Eastern. The frozen criteria, not interim score rankings, govern the experiment.

Frozen supervision specification

[9] bdd26. Next-stage research proposal: a new non-HoVer-Net model route. October 14, 2026. Internal design proposal; no PM-Net implementation or performance result. The present course proposal narrows its alternatives into a staged prototype plan.

Versioned new-model concept

Preparation note

Prepared against the Biomedical Data Design I Fall 2026 Project Proposal Guidelines: three aims; problem and motivation; data; prior work and baseline; computational approach; evaluation; deliverables; risks; scope and milestones; broader impact. Main proposal: eight pages including title and references. The separate contributions page follows.

Team contributions to this proposal

This page is separate from the eight-page main proposal.

The team will complete the statements below before submission.

TODO - COMPLETE BEFORE COURSE SUBMISSION
Add each team member's actual contribution to this proposal.

This section is intentionally left for the team to complete. Record work on the proposal itself, such as developing the research question, reviewing literature, designing experiments, preparing figures, or drafting and revising sections.

Team member	Contribution to this proposal
Zhiyu Song	**TODO: add actual contribution.**
Yuchen Ai	**TODO: add actual contribution.**
Xinhao Jin	**TODO: add actual contribution.**
Zheng Jian	**TODO: add actual contribution.**